# Implementation of a Fuzzy Inference System
# SATHYASEELAN C - 241001235

## Program:

```
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

temperature = np.arange(0, 41, 1)
fan_speed = np.arange(0, 11, 1)

temp_low = fuzz.trimf(temperature, [0, 0, 20])
temp_medium = fuzz.trimf(temperature, [10, 20, 30])
temp_high = fuzz.trimf(temperature, [20, 40, 40])

fan_low = fuzz.trimf(fan_speed, [0, 0, 5])
fan_medium = fuzz.trimf(fan_speed, [2, 5, 8])
fan_high = fuzz.trimf(fan_speed, [5, 10, 10])

plt.figure(figsize=(10, 6))

plt.subplot(2, 1, 1)
plt.plot(temperature, temp_low, label='Low')
plt.plot(temperature, temp_medium, label='Medium')
plt.plot(temperature, temp_high, label='High')
plt.title("Temperature Membership Functions")
plt.xlabel("Temperature (°C)")
plt.ylabel("Membership Degree")
plt.legend()

plt.subplot(2, 1, 2)
plt.plot(fan_speed, fan_low, label='Low')
plt.plot(fan_speed, fan_medium, label='Medium')
plt.plot(fan_speed, fan_high, label='High')
plt.title("Fan Speed Membership Functions")
plt.xlabel("Fan Speed")
plt.ylabel("Membership Degree")
plt.legend()
```

```
plt.tight_layout()
plt.show()

temperature_input = 28

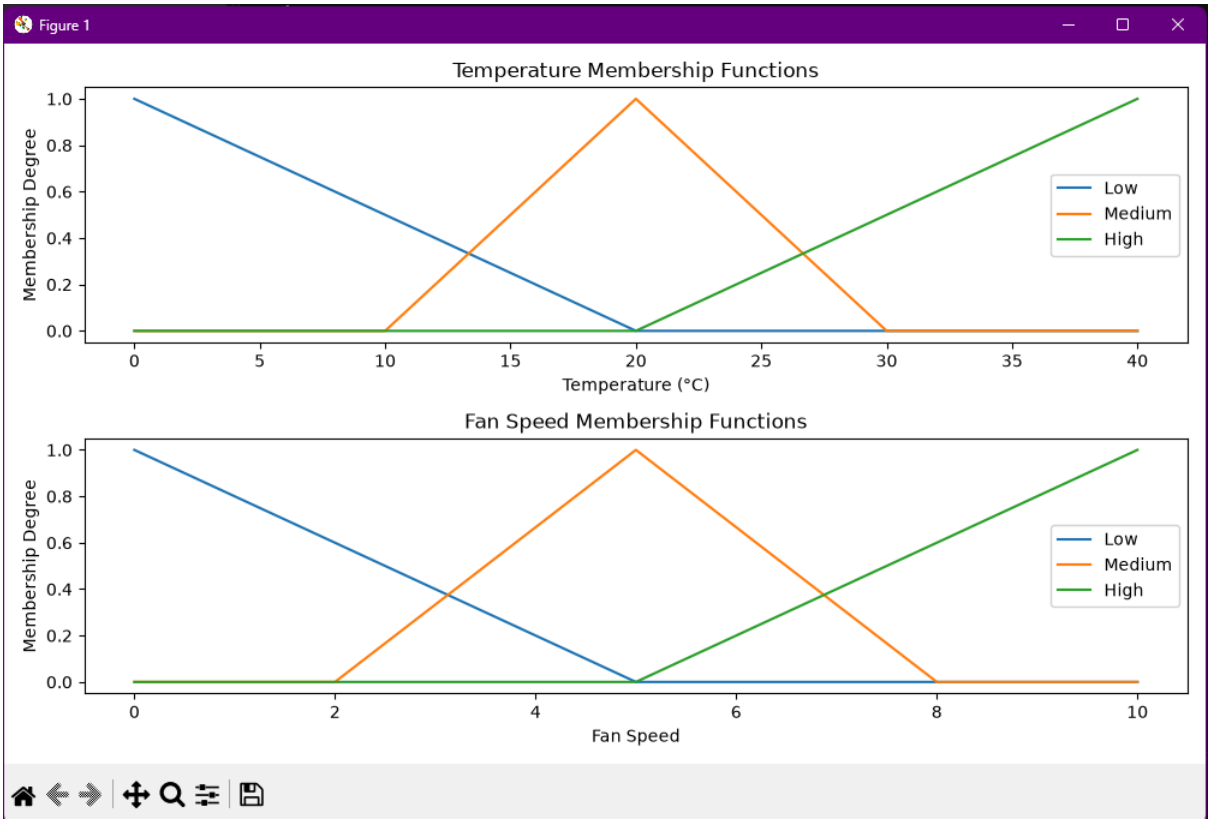
temp_low_level = fuzz.interp_membership(
    temperature, temp_low, temperature_input
)
temp_medium_level = fuzz.interp_membership(
    temperature, temp_medium, temperature_input
)
temp_high_level = fuzz.interp_membership(
    temperature, temp_high, temperature_input
)

fan_low_level = temp_low_level
fan_medium_level = temp_medium_level
fan_high_level = temp_high_level

fan_output = fuzz.defuzz(
    fan_speed,
    fan_low * fan_low_level
    + fan_medium * fan_medium_level
    + fan_high * fan_high_level,
    'centroid'
)

print(f"Temperature: {temperature_input}°C")
print(f"Fuzzified fan speed: {fan_output:.2f}")
```

# Output:


Figure 1
Temperature Membership Functions
Membership Degree
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
35
40
Temperature (°C)
Low
Medium
High
Fan Speed Membership Functions
Membership Degree
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
Fan Speed
Low
Medium
High